From Permit to Packet: Composing Agent Authority with Public Internet Measurement Evidence

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Abstract

An authenticated AI agent can issue a syntactically valid request that exceeds its mandate; an external service can create a valid object that differs from the request; a signed collector can omit evidence; and an internally consistent report can make a claim stronger than its observations support. These are composition failures: each component may pass its local check while the chain as a whole fails to justify the final claim. We study this problem in an agent-mediated RIPE Atlas workflow and model five non-compensable transitions: mandate to action, action to platform effect, effect to evidence, evidence to bounded claim, and artifact integrity. An ordered verifier returns the earliest invalid transition rather than a single undifferentiated signature result.

The evaluation combines three forms of evidence. First, an author-designed matched corpus contains 21 positive/negative pairs plus one clean baseline. All five verifier treatments accept 22/22 authorised controls. Across 21 designed faults, SHA-256-only and platform-native treatments each reject 1, a Permit-only treatment rejects 8, an offline composed treatment rejects 20, and a composed treatment with a logically separate witness rejects and correctly localises all 21. Second, a deterministic seven-family mutation sweep rejects and localises 448/448 generated variations. Third, a prospective action publishes a signed Permit and the digest of an exact 3,340-byte request before RIPE assigns eight measurement identifiers. Four explicit probes return all 32 scheduled one-off results; 32/32 pass narrow protocol checks at 848 result-derived credits under a 1,060-credit ceiling. A GitHub-hosted implementation subsequently passes 56 gates over both repository and production artifacts.

The results establish conformance only for the versioned corpus and bounded pilot. RIPE public API rows are not RIPE-signed statements; the witness shares Tyche administration and the RIPE upstream; and the work claims neither SCITT nor RATS conformance, population error rates, global availability, legal compliance, or independent audit. The contribution is an executable method for making authority-to-claim composition failures inspectable.

1. Introduction

Security accounts of autonomous or semi-autonomous agents often begin with an identity and end with an audit log. The difficult middle is frequently left implicit. A principal may allow an agent to perform one bounded action, but the agent sends a richer request. A service may accept that request yet create a different object. A collector may retrieve only part of the resulting state. Finally, a publisher may turn a local observation into an expansive statement. None of those failures is repaired by the fact that some JSON object was signed.

This distinction matters because cryptographic integrity, authorization, platform state, evidentiary coverage, and claim sufficiency answer different questions. A valid signature identifies control of a key over bytes. It does not itself show that the signer possessed relevant authority, that the bytes were sent, that the remote effect matches them, that all material results

were retained, or that the final statement follows from those results. Treating one check as a substitute for another produces what this paper calls *compensating verification*: a strong later artifact is incorrectly allowed to compensate for a broken earlier transition.

RIPE Atlas is a useful public setting in which to make the problem concrete. Users submit measurement definitions and source selections; RIPE Atlas assigns measurement identifiers and schedules probes; public APIs expose metadata and results; and a downstream analyst derives a claim. The platform therefore creates an externally assigned effect that does not exist when the principal authorises the action. Its credit model also gives authorization a measurable resource dimension. At the same time, the platform is not an attestation service for this experiment: its public rows are retrieved over TLS but are not operator-signed statements supplied for third-party offline verification.

The study asks four research questions:

- **RQ1 — separation:** Which designed failures are detected when a verifier checks only file integrity, only native platform state, only pre-execution authority, or the composed transition chain?
- **RQ2 — non-trivial acceptance:** Can the composed verifier reject the designed faults without achieving its score through blanket denial?
- **RQ3 — temporal binding:** Can a publicly committed Permit bind an exact request to platform identifiers that do not yet exist and later to a bounded evidence window?
- **RQ4 — reproducibility:** Can a verifier executed away from the action host reproduce the prospective verdict using public, privacy-minimised artifacts?

The paper makes four bounded contributions. It defines an ordered authority-to-claim transition model; provides an executable matched corpus and five verifier treatments; records a prospective RIPE Atlas trace whose Permit precedes the platform-assigned IDs; and publishes a production Observatory, tests, manifests, and an infrastructure-separated verification workflow. It does not claim that Permits, evidence receipts, first-failure diagnostics, or transparency mechanisms are new primitives. The narrower contribution is their composition around an external platform effect.

2. Background and related work

2.1 Delegated authority and attenuation

Delegation systems distinguish authenticated identity from the authority to perform a particular operation. SPKI/SDSI makes authorization certificates and delegation explicit [1]. Macaroons show how bearer credentials can be narrowed by contextual caveats concerning where, when, by whom, and for what purpose a request may be used [2]. These systems motivate the Permit in this experiment: the relevant object is not an assertion that an agent exists, but a bounded grant constraining target, action shape, execution host, source count, time, automatic replenishment, and cost.

The experiment does not implement SPKI or Macaroons. Its local Permit is a signed research object designed to isolate transition semantics. The important inheritance from authorization work is monotonicity: an execution may stay within or narrow its grant, but should not silently enlarge it. Positive controls therefore vary permitted values without breaking containment, while matched negatives cross one corresponding boundary.

2.2 Signed statements, provenance, and transparency

End-to-end provenance systems demonstrate the value of checking transitions instead of only final artifacts. In-toto records and verifies the steps of a software supply chain so that compromise at one stage cannot be hidden by a valid final package [3]. Certificate Transparency

uses append-only logs, inclusion evidence, and gossip to make conflicting views detectable [4]. Sigstore applies identity-bound signing and a public transparency log to software artifacts [5]. These systems address different objects, but share the principle that later integrity does not retroactively legitimate an earlier untrusted step.

The IETF SCITT architecture generalises signed statements and transparency services. A Signed Statement can be registered in a verifiable data structure; the resulting Receipt supports offline verification of inclusion, while the registration policy and append-only history support accountability [6]. SCITT also makes an important negative point: transparency does not prevent a dishonest or compromised issuer. This study uses that boundary directly. Its local Ed25519 envelopes are signed statements in the ordinary sense, but are not SCITT Signed Statements, COSE receipts, or entries in a SCITT transparency service.

2.3 Evidence appraisal and role separation

The RATS architecture distinguishes an Attester that produces Evidence, a Verifier that applies appraisal policy, and a Relying Party that uses Attestation Results for an application decision [7]. It also warns that when roles are aggregated in one entity, implementers must reassess the assumed security properties. This paper follows the role distinction but not a RATS wire format. The RIPE rows, collector statement, witness statement, and final claim are not represented as standards-conformant RATS conceptual messages.

Role separation is central to the strongest designed fault. If a collector can change an evidence index, recompute the summary, rewrite the claim, and sign the new internally consistent chain, an offline verifier with only the collector's view cannot detect equivocation. A witness with a prior semantic index can. However, logical separation is not compromise independence: both Tyche statements were produced on the same host, administered by the same organisation, and read the same RIPE upstream.

2.4 Emerging agent-action profiles

The SCITT Permit Internet-Draft binds a pre-execution authorization decision to canonical request bytes and pairs it with a post-dispatch Closure Record whose dispatch digest can be compared with the authorised digest [8]. The AI Agent Execution Profile Internet-Draft describes signed, hash-chained records of material agent actions and maps agent roles into SCITT [9]. These drafts are direct prior art for request digests, closure records, execution statements, and evidence chains. They are works in progress, not standards, and this prototype does not claim conformance to either.

The remaining seam is the platform join. A dispatch digest can show that the authorised bytes match recorded dispatch bytes, yet an external service may assign identifiers, normalise definitions, schedule only some sources, or produce state whose evidentiary meaning is narrower than a later claim. The present model makes those joins explicit and independently falsifiable.

2.5 RIPE Atlas as an external effect producer

RIPE Atlas is a global distributed Internet measurement system operated by the RIPE NCC. Probes receive measurement instructions through the platform and return results to its infrastructure [10,11]. User-defined measurements specify definitions, probe selection, and timing; supported types include ping, traceroute, DNS, TLS, and NTP [12]. Measurements consume credits and are assigned platform states such as Specified, Scheduled, Ongoing, and Stopped. Public measurement metadata and results can be retrieved through REST APIs [12,13].

Atlas therefore offers a real, inspectable platform effect without giving the experiment control of the assigned identifiers. It also exposes important limits. Probe selection is not a population sample, measurement success is protocol-specific, and public API retrieval is not equivalent

to an operator signature over the semantics used by this paper. The prospective claim is accordingly restricted to one window, four explicit probes, eight definitions, and narrow per-protocol success predicates.

3. Transition model

3.1 Artifacts and actors

The model contains six principal artifacts:

1. a **Permit** signed before action and containing authority limits;
2. an **action request** represented by exact canonical bytes and their digest;
3. a **platform effect** containing externally assigned measurement IDs and native definition/state fields;
4. an **evidence index** binding those IDs to captured metadata, results, source coverage, freshness, and derived counts;
5. a **claim** restricted to the facts and caveats supported by that evidence;
6. an **integrity manifest** binding retained public artifact bytes.

The principal authorises an operator-controlled agent. RIPE Atlas produces the external effect. A collector retrieves and summarises public platform data. A witness may retain a separately signed semantic view. A verifier appraises the chain, and a reader or automated relying party consumes the verdict.

3.2 Ordered, non-compensable gates

The verifier evaluates the following gates in order:

1. **Integrity:** retained components match the SHA-256 manifest.
2. **Mandate → action:** the Permit signature and validity interval pass; the exact request digest, target, definition set, sources, duration, execution host, automatic top-up rule, and credit ceiling are authorised.
3. **Native platform effect:** the external object has the required native structure and state.
4. **Action → effect:** assigned definitions, sources, identifiers, and time bounds correspond to the request.
5. **Effect → evidence:** the capture covers the assigned ID set and scheduled sources, falls inside the accepted time window, and derives its summary from the retained index.
6. **Evidence → claim:** quantitative facts equal the evidence and qualitative language remains a bounded observation with required caveats.
7. **Witness agreement:** when the witnessed treatment is selected, the collector and witness semantic indices agree.

On failure the verifier returns `DENY`, a machine-readable gate, and the first bad transition. Ordering is diagnostic, not compensatory: once an earlier edge fails, later validity cannot change the verdict. A multi-fault case tests that the earliest edge remains stable.

3.3 Threat model

The designed adversary may modify unsigned content, rehash modified content, substitute a valid platform object, omit evidence, alter a summary, overstate a claim, or control the collector's signing key. It may also create multiple faults. The adversary does not break Ed25519, SHA-256, TLS, GitHub, RIPE Atlas, or the host operating system. The witness treatment assumes the adversary does not simultaneously control both the collector and the witness view.

The prospective case is weaker than this abstract treatment assumption. The collector and witness use distinct statements and keys but share Tyche administration, the Zeus1 host and

network, and RIPE as upstream. The GitHub runner removes Zeus1 from repeated verification but remains under the same repository administration. These facts are reported as boundaries rather than treated as independence.

3.4 Cryptographic profile

Fixture and pilot envelopes use Ed25519 [14]. Structured values are serialized by recursively sorting object keys and emitting compact UTF-8 JSON. The exact API request is additionally retained as bytes, so its SHA-256 is computed over the actual 3,340-byte body rather than a later reconstruction. Canonicalization is required for repeatable hashing and signing [15], but the prototype's local serializer is not claimed to implement RFC 8785 JCS. Fixture private keys are deterministic and explicitly marked as test keys; pilot private keys remain outside the public artifact.

4. Study design and implementation

4.1 Five verifier treatments

Five treatments isolate the information available at appraisal time:

- **SHA-256 only** checks the retained artifact manifest.
- **Platform native only** checks the expected RIPE object structure/state.
- **Permit only** verifies the Permit and mandate-to-action containment.
- **Composed offline** checks integrity and every local transition through the bounded claim.
- **Composed + witness** adds agreement with a separately signed semantic evidence index.

The treatments are intentionally not competitors of equal scope. They expose which semantic classes each information boundary cannot observe.

4.2 Matched replay corpus

The replay corpus contains 43 bundles: 21 designed negative cases, 21 matched authorised controls, and one unpaired clean baseline. Each pair varies the same semantic dimension on opposite sides of its policy boundary. For example, a negative target substitution changes the target outside the Permit, whereas its positive twin changes target and Permit consistently; a negative evidence omission removes an assigned measurement, whereas its positive twin only reorders a complete set. This structure prevents a verifier that rejects every variation from appearing safe.

The faults cover target substitution, probe-budget and duration escalation, automatic replenishment, required-definition removal, execution-host escape, invalid Permit signature, request/effect mismatch, platform-definition and time substitution, measurement omission, summary cherry-picking, stale capture, three forms of semantic overclaim, missing boundaries, platform state failure, unrehashed integrity tampering, collector reissue, and a multi-fault first-edge case.

Expected ground truth and expected first bad edge are authored with the cases. The corpus measures conformance to those authored expectations. It does not estimate how often such failures occur in deployed agents or RIPE Atlas.

4.3 Generated variations

A seeded generator (`20260813`) creates 64 variants in each of seven operator families: target substitution, probe-budget escalation, duration escalation, execution-host escape, platform-effect substitution, evidence-summary cherry-picking, and claim overreach. The 448 cases

broaden input coverage around known fault operators but are not an independent adversarial sample.

The first developmental sweep exposed a layer-placement defect: the native platform treatment incorrectly examined a local evidence summary. The check was moved to effect-to-evidence before the reported corpus and sweep were frozen. This history is disclosed because the generator informed implementation correction; it is not a preregistered confirmatory result.

4.4 Prospective protocol

Before submission to RIPE Atlas, the experiment published the exact canonical request, SHA-256 digest, public experiment key, signed Permit, authority limits, and an empty measurement-ID list. The request contained eight public one-off definitions for `tyche.institute`: IPv4 and IPv6 variants of ping, traceroute, DNS, and TLS certificate retrieval. It named probes 28, 81, 6349, and 6569, disabled automatic top-up, authorised only `zeus1`, and capped expenditure at 1,060 credits.

The action was sent through an authenticated Chrome session on Zeus1. The initial session-authenticated attempt returned HTTP 403 before creation because the RIPE endpoint required `X-Requested-With`. Adding that transport header did not change the committed JSON body. The accepted response then assigned eight measurement IDs.

The collector retrieved public metadata and results, retained a private raw capture on Zeus1, and published a privacy-minimised source index and file-hash manifest. A Chrome-CDP replay used separate collection code and a separate key to compare measurement IDs, counts, narrow successes, and the semantic source index. Raw rows containing network-address fields are excluded from the public repository.

4.5 Outcome predicates

For the prospective pilot, success is deliberately narrow. A ping row passes when all requested replies are represented; traceroute passes when the destination response is reached; DNS passes when the requested answer is returned; and TLS passes when a certificate chain is returned. These checks do not establish service availability, path quality, certificate policy compliance, or security. Credit usage is derived from RIPE per-result metadata and checked against the Permit ceiling.

4.6 External reproduction

A Node implementation in GitHub Actions verifies Ed25519 envelopes, the exact request digest, target and source constraints, disabled top-up, credit ceiling, platform ID identity, claim counts, and public capture-manifest hashes. It runs against both checked-out repository artifacts and production URLs. Its 56 gates are a second implementation of the prospective appraisal path, not of every Python replay-corpus treatment.

4.7 Metrics

For matched cases, the study reports authorised acceptance, designed-fault detection, balanced accuracy, and correct first-edge localisation. Balanced accuracy is the mean of the authorised acceptance rate and negative detection rate. Because the corpus is fully authored and small, no confidence intervals or population inference are reported. Generated variations are reported as exact seeded coverage. Performance uses `perf_counter_ns`; each observation is a full 43-case pass divided by case count, repeated 2,000 times on the recorded Zeus1 software environment.

5. Results

5.1 Matched corpus

Treatment	Authorised accepted	Faults denied	Balanced accuracy	Correct first-edge locations
SHA-256 only	22/22	1/21	0.524	1/21
Platform native only	22/22	1/21	0.524	1/21
Permit only	22/22	8/21	0.690	8/21
Composed offline	22/22	20/21	0.976	20/21
Composed + witness	22/22	21/21	1.000	21/21

All treatments accept all authorised controls, answering RQ2 within the corpus: the composed result is not produced by blanket denial. The hash-only treatment detects unrehashed tampering but accepts 20 semantically changed faults after their artifacts are made internally hash-consistent. The native treatment detects the stopped-platform case but has no authority or claim context. The Permit treatment detects all eight mandate-to-action faults and accepts 13 later-chain faults.

The composed offline treatment rejects and correctly locates 20 faults. Its sole false accept is `D20_collector_reissued_cherry_pick`: the collector removes part of the evidence index, recomputes the summary and claim, and re-signs the consistent chain. Nothing in that collector-controlled view contradicts itself. The witnessed treatment compares the prior semantic index and returns `evidence.witness_disagreement`, detecting the 21st fault. This is evidence for the value of a separate view, conditional on its uncompromised and non-colluding status.

5.2 Generated sweep

The witnessed composed verifier rejects 448/448 generated variations and reports the expected first bad edge in 448/448. Each of the seven families contributes 64 cases. The exact result answers an implementation-coverage question only; it is not an empirical estimate of detection against unknown attacks.

5.3 Prospective trace

The public precommit digest was:

`ae3c95459fba6e8a3fb1120aeecf47529a214aaa0bfacceb45ee0a7d051c91ad`

The accepted RIPE response assigned IDs `200503603`, `200503604`, and `200503606–200503611`. Each of the four requested probes was scheduled in each definition. The capture retained all 32 expected rows, and 32/32 passed the narrow protocol predicates. RIPE per-result metadata accounts for 848 credits, below the 1,060-credit Permit ceiling.

The collector Closure verifies and returns `ALLOW` with no first bad edge. The Chrome-CDP witness agrees on identifiers, result count, success count, and source index. The prospective trace therefore answers RQ3 for this action: the pre-action request digest is joined to subsequently assigned platform IDs and then to a complete bounded evidence index. It does not prove that the public RIPE semantics themselves are signed or independently witnessed.

5.4 Infrastructure-separated verification

The GitHub-hosted runner passes all 56 gates over both its checkout and the public production URLs. Both paths reproduce the request digest, eight IDs, 32 rows, 32 narrow successes, and 848 credits. This answers RQ4 in the limited sense that appraisal no longer depends on executing the verifier on Zeus1. It does not create an independent auditor because Tyche controls the repository, workflow, and public presentation.

5.5 Cost

Treatment	Median µs/bundle	p95 µs/bundle
Platform native only	0.957	1.008
SHA-256 only	51.592	54.160
Permit only	85.596	87.779
Composed offline	208.974	219.822
Composed + witness	273.444	362.659

The clean replay bundle is 11,341 bytes, the signed prospective Closure is 10,309 bytes, and the 17 retained private capture JSON files total 80,469 bytes. These are descriptive results from one Python 3.12.3 / `cryptography` 49.0.0 environment on Linux x86-64, not cross-machine performance claims.

6. Discussion

6.1 What the treatment differences mean

The treatment comparison makes the semantics of common assurance shortcuts visible. A manifest can establish that retained bytes equal a published digest; it cannot decide whether a rehashed request was authorised. A platform-native check can establish that an object is structurally plausible and in an expected state; it cannot know what a principal allowed. A Permit can prevent scope escalation; it cannot establish that later evidence is complete. A complete evidence index still cannot justify language broader than its window.

The implication is not that every application needs seven gates. It is that a relying party should name which transition supports which claim. An architecture that stops after signature verification should make the resulting unknowns explicit rather than inheriting authority or truth from the signature.

6.2 First-bad-edge diagnostics

A Boolean result is often insufficient for repair. If a substituted target and an incomplete capture occur together, fixing the capture does not make the action authorised. Ordered first-bad-edge reporting makes this precedence explicit. It also prevents later checks from obscuring the earliest policy failure. The order is application policy, however, not a universal theorem; other workflows may place native safety gates before authority or distinguish more granular evidence stages.

6.3 Temporal evidence and externally assigned identifiers

The prospective trace supplies evidence unavailable in a retrospective fixture: the RIPE IDs are absent when the Permit is published. Their later appearance shows an actual temporal join from authorised request bytes to an externally created object. This is more informative

than inserting known IDs into a signed record after the fact, while remaining weaker than an operator-issued receipt that cryptographically binds the request and resulting effect.

The clean next standardisation path would keep the semantic gates while replacing local envelopes with standards-based signed statements and receipts. A SCITT Permit and Closure could bind authorisation and dispatch; a transparency service could provide inclusion and non-equivocation evidence; a RATS appraisal could cover execution-platform trust where relevant; and a domain adapter could bind the external platform effect and evidence semantics. The present artifact tests the joins without claiming those standards have been implemented.

6.4 Independence as a graded property

The D20 result demonstrates why a separately signed statement is not enough if its producer shares the collector's compromise domain. Independence has at least four dimensions: key separation, software-path separation, administrative separation, and upstream separation. The pilot achieves the first two only partially and the GitHub run adds execution-host separation. Future work should place the witness under a different administrator and network, retain a prior commitment in an append-only service, and compare more than one upstream representation where the platform permits it.

6.5 Thesis and publication value

For a thesis on agent authority or verifiable action evidence, the project serves as a worked empirical bridge between abstract delegation and observable external effects. Its value is methodological: it shows how to turn a claim of authority propagation into falsifiable joins, matched controls, typed failure attribution, a prospective trace, and a reproducible artifact. It should not be used as evidence that RIPE Atlas itself is unreliable or that one witness architecture is sufficient for deployment.

7. Threats to validity

Construct validity. The authored gates operationalise one model of authority-to-claim composition. Other applications may require different semantics or ordering. The narrow protocol predicates are intentionally weaker than availability or security claims.

Internal validity. The same author designed the corpus and the main Python verifier. Matched controls reduce the risk of trivial rejection, and the Node prospective verifier reduces implementation-path dependence, but neither constitutes independent ground-truth annotation. The mutation generator helped find and correct a gate-placement defect before the final run.

External validity. The corpus has 21 designed faults and 22 authorised controls. The live trace is one action over four convenience-selected probes and one target. No prevalence, ecosystem coverage, or RIPE reliability conclusion follows.

Evidence validity. RIPE Atlas public API rows are upstream data retrieved over TLS, not RIPE-signed evidence in this artifact. The collector and witness share an upstream, host, network, and administrator. The GitHub runner is infrastructure-separated but not institutionally independent.

Temporal validity. Internet state, probe connectivity, software versions, standards drafts, and public endpoints change. Results are version- and window-bound. The eight one-off measurements are stopped historical objects; they are not a continuing monitor.

8. Ethics, privacy, and responsible use

The pilot uses public RIPE Atlas measurements with automatic top-up disabled and an explicit credit ceiling. It targets the Institute's own public domain and performs ordinary ping, traceroute, DNS, and TLS-certificate measurements. The public artifact excludes authenticated API credentials, private keys, and raw address-bearing result rows. The raw capture is retained privately on Zeus1; a privacy-minimised manifest exposes filenames, sizes, and SHA-256 digests for integrity checking.

The Observatory labels the work as a research prototype and states prohibited inferences. A valid bundle must not be presented as proof of legal authority, human identity, compliance, RIPE NCC membership, LIR status, general security, or global availability. Researchers reusing the software remain responsible for platform terms, measurement ethics, target permission, privacy, and credit budgets.

9. Reproducibility and availability

The public release contains source code, deterministic fixture keys, the 43 case bundles, case-level verdicts, generated-sweep results, performance metadata, prospective precommit and Closure artifacts, the privacy-minimised capture manifest, tests, and SHA-256 manifests. Reproduction requires Python 3.11 or later and `cryptography`. The core commands rebuild the corpus and results, execute the mutation sweep and microbenchmark, run five unit tests, and verify the file manifest.

The public Observatory is available at https://tyche.institute/lab/from-permit-to-packet/. The release repository is https://github.com/tyche-institute/from-permit-to-packet; the archival release is https://doi.org/10.5281/zenodo.21923744. Code is licensed under Apache License 2.0; the manuscript, documentation, and public data are licensed under CC BY 4.0. Private raw capture and private keys are not part of the release.

10. Conclusion

The study supports a modest but useful result: explicit, ordered transition gates can distinguish integrity, authority, native platform effect, evidence coverage, and claim sufficiency in a real agent-mediated public measurement workflow. The matched corpus shows why no single local treatment observes the whole chain; the prospective trace demonstrates temporal binding to platform identifiers that did not exist at authorisation time; and public verification shows that the bounded appraisal can be repeated away from the action host.

The decisive next work is institutional rather than cosmetic: independent labeling, a separately administered witness, cross-platform replication, property-based request generation, and standards-based receipts. Until then, the artifact is a reproducible research profile and thesis-grade pilot, not a production assurance service or certification mechanism.

Declarations

Author contributions. Anton Sokolov conceived the study, designed the artifact and corpus, operated the prospective pilot, analysed the results, and wrote the manuscript.

Generative-AI assistance. OpenAI Codex was used under author direction for software implementation, test generation, data checking, manuscript drafting, and editorial revision. The author selected the research questions, authorised the live action, reviewed the evidence boundaries, and remains responsible for the work and its claims.

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Competing interests. The author is affiliated with Tyche Institute, which operates the repository, Observatory, collector, witness, and verification workflow described here. No claim of institutional independence is made.

Data and code availability. Public, privacy-minimised evidence and code are released through the repository and archival DOI identified in Section 9. Raw network-address-bearing capture data and secret keys are not publicly released.

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